

Product/Business Requirements Document (PRD/BRD)

Intelligent Beat Plan / Permanent Journey Plan (PJP) Recommendation Engine

Field	Detail
Document Type	Combined PRD + BRD
Version	1.0 (Draft)
Prototype Market	Mumbai
Prepared For	Internal project use
Status	Draft — pre-development

1. Executive Summary

This document defines the business rationale, functional scope, and technical design for a **Beat Plan / Permanent Journey Plan (PJP) Recommendation Engine** — a system that generates and continuously optimizes field sales visit schedules. The system recommends **which outlets a rep should visit, in what order, and how often**, to maximize territory coverage and revenue while minimizing travel time and enforcing visit discipline.

The initial build is a **single-industry, single-geography prototype** (FMCG/General Trade, Mumbai) intended to validate the core algorithms — territory design and route optimization — before generalizing to other industries or cities.

2. Business Requirements (BRD Section)

2.1 Business Problem

Field sales teams in India typically plan beats manually or via static spreadsheets. This leads to:

Problem	Business Impact
Inconsistent outlet coverage	Revenue leakage from missed/under-served outlets
Inefficient route sequencing	Wasted travel time reduces number of productive visits/day
No visit discipline enforcement	Reps deviate from plan with no visibility or accountability
Static, one-time territory design	Territories don't adapt as outlet universe or demand changes
No data-driven prioritization	High-potential outlets get the same visit frequency as low-potential ones

2.2 Business Objectives

Objective	Success Metric	Target (Pilot Phase)
Increase outlet coverage per cycle	% of mapped outlets visited within SLA window	≥ 90%
Reduce non-productive travel time	Avg. travel time per rep per day	Reduce by 15-20% vs. manual baseline
Improve visit discipline	Beat adherence % (planned vs. actual)	≥ 85%
Improve revenue per beat	Sales per beat, before vs. after	Directionally positive (measured post-pilot)
Improve strike rate	% of visits resulting in an order	Establish baseline, then improve

2.3 Stakeholders

Role	Interest
Sales Rep (end user, field)	Simple daily plan, minimal manual effort, fair workload
Sales Manager / ASM	Visibility into adherence, coverage, override capability
Business/Category Head	Revenue impact, ROI on the tool
Data/Analytics Team	Data pipeline health, model performance
IT/Engineering	System reliability, integration with existing ERP/DMS

2.4 Business Scope

In Scope (Phase 1 — Prototype) - Single industry vertical: FMCG General Trade - Single geography: Mumbai (test area first, expandable) - Outlet universe sourced from open data (OpenStreetMap) + manually curated additions - Synthetic rep/visit/order data for algorithm validation (no live pilot yet)

Out of Scope (Phase 1) - Multi-industry support (Pharma, Industrial, B2B — deferred to later phase) - Live ERP/DMS integration - Real-time GPS tracking and geofencing enforcement - Mobile app for reps (prototype is backend/algorithm-first) - SKU-level recommendation ("next-best product") — deferred to Phase 3

2.5 Assumptions

- Outlet data from OpenStreetMap is reasonably accurate for Mumbai but incomplete; manual supplementation will be needed for a real pilot.
- No historical visit/order data exists; synthetic data will be used to validate algorithm logic, not business impact.
- Road travel time (not straight-line distance) is required for realistic routing given Mumbai traffic patterns.
- A single rep's daily capacity (working hours, break time) is configurable but assumed fixed per rep for prototype purposes.

2.6 Dependencies

- Availability of OSM data density in the chosen test area.
- Access to a routing engine (OSRM self-hosted, or Google Maps API for higher accuracy).
- Future dependency (post-prototype): access to real ERP/DMS order and outlet data for validation against actual business impact.

2.7 Risks

Risk	Likelihood	Impact	Mitigation
OSM data sparse/inaccurate in parts of Mumbai	Medium	Medium	Manual spot-check and supplement key areas
No real data means algorithm validated on synthetic data only	High	Medium	Be explicit that Phase 1 output is algorithmic proof-of-concept, not business-validated
Route optimization doesn't scale well as outlet count grows	Medium	High	Use heuristic solvers (OR-Tools) designed for scale from the start
Stakeholders expect immediate ROI without a live pilot	Medium	Medium	Set expectations clearly: Phase 1 = proof of concept, not deployment

3. Functional Requirements (PRD Section)

3.1 User Roles

Role	Description	Access Level
Field Sales Rep	Views assigned daily/weekly beat plan	Read-only (own plan)
Sales Manager	Views team's plans, adherence, can approve beat changes	Read + approve
System/Admin	Configures outlet master, tiering rules, algorithm parameters	Full

(Note: Phase 1 prototype has no live UI for reps/managers — these roles define the target functional model for when the system moves beyond prototype.)

3.2 Functional Modules & Requirements

FR-1: Outlet Master Management

ID	Requirement
FR-1.1	System shall ingest outlet data (name, type, lat/long, address) from OpenStreetMap Overpass API for a defined bounding box
FR-1.2	System shall allow manual addition/edit/deactivation of outlets
FR-1.3	System shall classify each outlet into a tier (A/B/C) based on configurable rules (initially: assumed potential; later: historical revenue)
FR-1.4	System shall assign a default visit frequency norm per outlet type (configurable)

FR-2: Territory / Beat Design

ID	Requirement
FR-2.1	System shall cluster outlets into geographically coherent beats using a clustering algorithm (e.g., k-means or DBSCAN on lat/long)
FR-2.2	System shall balance beats by rep capacity (max outlets/day × cycle length)
FR-2.3	System shall allow manual override of auto-generated beat boundaries
FR-2.4	System shall support beat redesign on demand (e.g., when outlet universe changes)

FR-3: Route Optimization (Daily Sequencing)

ID	Requirement
FR-3.1	System shall generate an optimized visit sequence for a rep's daily outlet list, minimizing total travel time
FR-3.2	System shall use actual road travel time (not straight-line distance) via a routing engine

FR-3.3	System shall respect rep constraints: working hours, break duration, per-outlet visit duration
FR-3.4	System shall flag infeasible plans (e.g., more outlets than time allows) rather than silently truncating

FR-4: Visit Recommendation Logic (Phase 1: rule-based; Phase 3: ML-driven)

ID	Requirement
FR-4.1	System shall recommend visit frequency per outlet based on tier and outlet type
FR-4.2	System shall (Phase 3) adjust frequency dynamically based on order history/behavior patterns
FR-4.3	System shall (Phase 3) recommend outlet/SKU focus areas for a visit — deferred

FR-5: Compliance & Tracking (Post-Prototype)

ID	Requirement
FR-5.1	System shall capture check-in/check-out GPS and timestamp per visit
FR-5.2	System shall validate check-in against a configurable geofence radius
FR-5.3	System shall log deviation reason codes for skipped/rescheduled visits

FR-6: Analytics & Reporting

ID	Requirement
FR-6.1	System shall compute and display coverage % (visited/mapped outlets) per cycle
FR-6.2	System shall compute beat adherence % (planned vs. actual visits)
FR-6.3	System shall compute strike rate (% of visits with an order) once order data exists
FR-6.4	System shall visualize beats and routes on a map (outlet points, route lines, beat boundaries)

3.3 Non-Functional Requirements

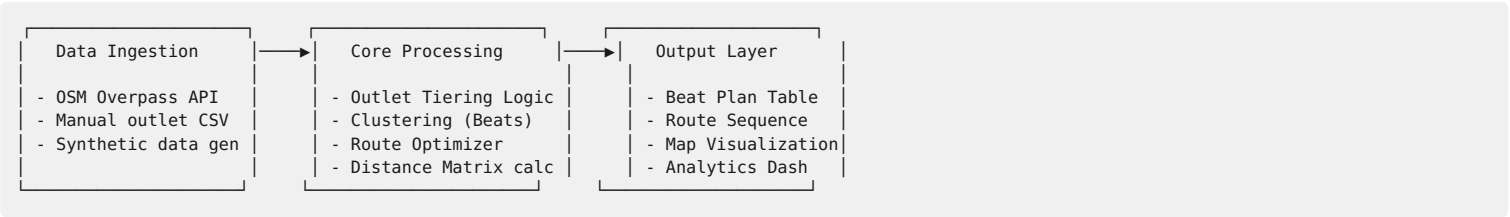
Category	Requirement
Performance	Route optimization for a single rep's daily outlets (~30-50 stops) should complete in under 5 seconds
Scalability	Territory design should handle at least 5,000-10,000 outlets for a city-scale rollout without redesign of core architecture
Data Freshness	Distance/travel time matrix should be refreshable (traffic conditions change; not a one-time static cache)
Usability	Output (beat plans, maps) should be consumable without requiring the end user to understand the underlying algorithm
Portability	Core logic should not be hard-coded to FMCG — data model must support other industry verticals (per original schema design)
Offline Resilience	<i>(Post-prototype, mobile app phase)</i> — rep-facing app must function with intermittent connectivity, typical of Tier-2/3 India

3.4 Out-of-Scope Functional Items (Explicitly Deferred)

- Rep-facing mobile application (UI/UX)
- Real-time GPS tracking during transit (only check-in/out points, not live tracking)
- Multi-language support
- Integration with specific ERP systems (SAP, Dynamics, etc.) — deferred until a real pilot partner is identified
- Incentive/commission calculation tied to visit performance

4. Technical Design (Technical PRD Section)

4.1 High-Level Architecture



4.2 Technology Stack (Proposed)

Layer	Technology Options	Recommendation for Prototype
Data Ingestion	Overpass API, CSV import	Overpass API (direct HTTP calls)
Data Storage	PostgreSQL + PostGIS, or flat files for prototype	PostgreSQL + PostGIS (geo-indexing support)
Clustering (Territory Design)	k-means, DBSCAN, or capacity-constrained clustering (scikit-learn)	DBSCAN or capacity-constrained k-means

Route Optimization	Google OR-Tools (VRP solver), OSRM for distance	Google OR-Tools + OSRM (both free/self-hosted)
Distance/Travel Time	OSRM (self-hosted), Google Distance Matrix API (paid)	OSRM for prototype (free); evaluate Google API for pilot accuracy
Backend	Python (FastAPI/Flask)	Python — aligns with OR-Tools, scikit-learn, geo libraries
Visualization	Folium/Leaflet (maps), Plotly/Dash or a simple web frontend	Folium for quick map prototyping
Analytics Dashboard	Streamlit (fast prototype), or Power BI if enterprise-preferred	Streamlit for Phase 1 prototype

4.3 Data Architecture

Refer to the separate **Data Schema document** (`pjp-data-schema.md`) for full entity definitions. Summary of core entities:

Entity	Source (Phase 1)
Outlet Master	OpenStreetMap Overpass API + manual curation
Rep Master	Synthetic (generated for prototype)
Beat/Territory Master	Generated by clustering algorithm
Beat Plan	Generated by route optimizer
Visit Log	Synthetic (until real pilot)
Orders	Synthetic (until real pilot)
Distance Matrix	Computed via OSRM

4.4 Core Algorithms

Function	Algorithm/Approach	Notes
Territory Clustering	Capacity-constrained clustering (e.g., k-means with post-hoc rebalancing, or DBSCAN + manual capacity split)	Must respect max-outlets-per-rep constraint, not just geographic proximity
Route Sequencing	Vehicle Routing Problem (VRP) via OR-Tools; TSP as a simplified single-rep case	OR-Tools handles time windows and capacity constraints natively
Outlet Tiering	Rule-based scoring (Phase 1): weighted score from outlet type + assumed potential	Move to data-driven scoring (historical revenue) once real data exists
Visit Frequency Recommendation	Static lookup table by tier (Phase 1)	ML-driven dynamic frequency deferred to Phase 3

4.5 Integration Points (Future — Not Phase 1)

System	Purpose	Priority
ERP/DMS (e.g., MS Dynamics, SAP)	Real order and outlet data	High (post-prototype)
Maps/Routing API (Google)	Higher-accuracy travel time	Medium
Mobile App Backend	Push daily plans, receive check-in/out events	High (post-prototype)

4.6 Technical Constraints

- Straight-line (haversine) distance is **not acceptable** for route sequencing in Mumbai — road network routing (OSRM/Google) is mandatory.
- OSM data quality varies by area — technical validation step needed (spot-check density/accuracy before trusting a cluster's outlet count).
- Route optimization must be re-run when outlet universe changes — not a one-time static computation.

5. Success Criteria for Phase 1 (Prototype Exit Criteria)

Criterion	Definition of Done
Outlet data pulled	Outlet Master populated for one Mumbai test area via OSM
Beats generated	Outlets successfully clustered into capacity-balanced beats
Routes generated	Daily visit sequence generated per synthetic rep, using real road travel times
Visualization	Beats and routes visible on a map
Baseline metrics	Coverage %, adherence % (using synthetic visit data) computed and displayed

Phase 1 is explicitly a **proof-of-concept on real outlet geography with synthetic behavioral data** — it validates the algorithms, not business ROI. Business impact validation requires a live pilot with a real sales team, which is Phase 2+.

6. Phased Roadmap (Cross-Reference)

Phase	Deliverable
1	Outlet data + territory clustering + route optimization (this PRD's scope)
2	Live pilot: real rep, real visits, mobile check-in, compliance tracking
3	SKU-level recommendation engine, dynamic visit frequency (ML-driven)
4	Multi-industry, multi-city generalization

7. Open Questions (To Be Resolved Before Development Starts)

- Which specific Mumbai test area/pincode should be used for the prototype?
- What is the assumed rep capacity (working hours, avg visit duration) for synthetic data generation?
- Should Phase 1 use OR-Tools (self-hosted, free) or evaluate a paid solver for better performance at scale?
- Is there a target internal stakeholder or team this prototype will eventually be demoed to, which might shape what "success" needs to look like?